# VAIBHAV ITAURIYA

Final Year Undergraduate
Department of Mechanical Engineering
Indian Institute of Technology Kanpur

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| Examination | Degree/Certificate | Institute | Year | CPI/% |
|---|---|---|---|---|
| Graduation | B.Tech | Indian Institute of Technology Kanpur | 2023-2027 | 7.4/10 |
| Intermediate/+2 | CBSE (XII) | The Gramodaya International College, Mauranipur | 2023 | 91.4% |
| Matriculation | CBSE (X) | The Gramodaya International College, Mauranipur | 2021 | 94.8% |

## SCHOLASTIC ACHIEVEMENTS

- Recipient of the **National Talent Search Examination (NTSE) Scholarship**, awarded by **NCERT, Government of India** in **2021**
- Completed **SURGE 2025** Research Internship at IIT Kanpur, selected among the **top 300** out of **3000+** applicants across India
- Secured **956th International Rank** in **SOF Science Olympiad 2021**, among **2.4 Million+ participants** spanning 70+ countries

## WORK EXPERIENCE

**Jaguar Land Rover (JLR TBSI), Bengaluru** *May'26 - Jul'26*
*Software Intern*
*Won **1st Place** in the **JLR Global Hackathon** across **7+ countries***

- Built an **engineering lifecycle traceability platform** unifying Requirements, Design, Code and Tests across **200+ repositories**, used by **10+ global teams**, **150+ engineers** and **5+ JLR auditors**
- Engineered a **Git-diff based pipeline** using GitLab CI, Python, Lambda and Redshift that parses **20+** fields per artifact with upsert-by-ID sync, cutting repo sync time from **2 min to 5 sec**
- Designed **8 filter-aware API routes** using Lambda, API Gateway and Redshift; migrated data layer from **DynamoDB** to **Redshift**
- Built a **Next.js** and **TypeScript** dashboard with Tailwind, Recharts and React Flow for artifact filtering, trace graphs and mapping
- Cut cached API latency from **3 sec to 2 ms** with **Redis caching** and page load time from **20 sec to 2 sec** using server-side pagination
- Developed **BFS-based traceability engine** for **50K-node graphs**, with filter-aware, depth-controlled traversal and GitLab file links
- Automated traceability checks, replacing a **3-week manual audit workflow** with on-demand analysis across **1 Million+ artifacts**

**PFA India, *Personal Finance Startup*** *Jan'25 - Aug'25*
*Product Management Intern*
*Internship extended by **2 months** based on **strong performance***

- Built a weekly **Python, Selenium** and **Pandas** web-scraping pipeline for **15+ banks**, scaling from **3 to 50+ webpages per day**
- Developed a **Python recommendation engine** scoring **25+ credit cards** on **150+ parameters**, suggesting the best-fit cards to users
- Built responsive **React frontends** for credit-card comparison and FIRE tools, translating workflows into intuitive user interfaces
- Drove **23-minute average sessions** and **8.4/10** satisfaction for card comparison and **7.5/10** for FIRE across all **200+** pilot users

## TECHNICAL ACHIEVEMENTS

- Secured **4$^{th}$ rank nationally** in Code & Compete at Techkriti'26 among **5,000+ participants** in **Data Structures and Algorithms**
- Ranked **254$^{th}$** among **9,000+** in **Goldman Sachs Hackathon'25**
- **Top 10** nationally in KUKU FM's **Project K** - NLP/RAG challenge

## POSITION OF RESPONSIBILITY

**Overall Placement Coordinator** *Mar'26 - Present*
*Students' Placement Office, IIT Kanpur*

- Spearheading a 4-tier team of **250+ students** to manage and conduct the Internship and Placement Drive for **3000+ students**
- Streamlined recruitment as Assistant Coordinator helping secure **1300+ placement offers** and **600+ internship offers** in 2025-26
- Onboarded new firms including **LinkedIn, PocketFM, Point72, BNP Paribas** and **Irage Broking** for the 2026-27 recruitment cycle

## PROJECTS

**Pacto: Agentic AI Accountability Agent** *Aug'26 - Oct'26*
*The Ken × Zerodha Case Competition 2026, National Finalist (Build Track)*

- Built an **agentic AI system** (**GPT-4o-mini**) on an **MCP server** with **39 tools**, verifying missed sessions and auto-settling user stakes
- Designed a **payment state machine** for charges, pre-debit notices, **Pine Labs** links and payouts with **idempotent API retries**
- Integrated **Gnani STT/TTS** for Hinglish Telegram voice notes, with server guardrails so money moves only on confirmed state
- Ran **18 end-to-end test rounds** (403s, 504 timeouts, duplicate charges), adding session-ID normalization and **charge blocking**
- Wired **Google Calendar** triggers, **GPS geofence** check-ins and **Gmail** receipts, with a scheduler running Pacto every **5 minutes**

**Amazon ML Challenge 2026** *Sep'26*
*Team Lead | 72-hour national ML hackathon by Amazon*

- Led a 4-member team building an end-to-end **entity-resolution pipeline** in **Python**, mapping noisy records onto **1.77M entities**
- Wrote **NLP preprocessing** for Indic transliteration and address normalization, and **TF-IDF top-k blocking** to speed up retrieval
- Engineered **75 pairwise features** (fuzzy matching, set overlaps) for a **two-stage LightGBM** classifier with isotonic recalibration
- Fine-tuned a **multilingual-e5 cross-encoder** (Hugging Face) to re-rank candidates, optimizing match thresholds for **macro $F_{0.5}$**
- Scored **0.9867 macro $F_{0.5}$** on the final leaderboard, ranking **384th among 25K+ teams** nationwide (**top 1.5%**) within just **72 hours**

**LoreDrop** *Jan'26 - Apr'26*
*CS253 Course Project, under Prof. Indranil Saha, CSE, IIT Kanpur*

- Built a campus event platform replacing scattered **WhatsApp and email updates** for all of IIT Kanpur's **47** registered student clubs
- Implemented **Firebase OTP authentication** and secure token based login, restricting platform access to verified IITK members
- Designed **role-based access control** for students, organizers and admins, ensuring only approved organizers could publish events
- Developed a personalized event feed with indexed **MongoDB** queries using **React**, **TypeScript** and **Node.js** for faster loading
- Added **scheduling clash detection**, recurring events and email alerts while collaborating in a **9-member team** using Git/GitHub

**LLuminating Language: RAG Chatbot** *May'24 - Jul'24*
*Brain and Cognitive Society, IIT Kanpur | **Best Implementation Award***

- Developed an IITK-specific **RAG chatbot** on a locally hosted **LLaMA3-8B** with LangChain and a **ChromaDB** vector database
- Built an end-to-end **RAG pipeline** with Selenium and Instructor embeddings, ingesting **400+ webpages** into **3000+ text chunks**
- Optimized **top-k semantic retrieval**, cutting query retrieval latency to **sub-200 ms** from an unoptimized baseline of **1.5 sec**
- Added **confidence-based guardrails** and prompt engineering to curb hallucinations, achieving **90%+ accuracy** on 200+ queries

## TECHNICAL SKILLS

- **Languages:** C/C++, Python, SQL, TypeScript, JavaScript, HTML
- **Web & Backend:** Next.js, React.js, Node.js, Django, RESTful APIs
- **Cloud & DevOps:** AWS (Lambda, API Gateway), Docker, CI/CD
- **Databases:** Redshift, DynamoDB, MongoDB, Redis, ChromaDB
- **AI & ML:** LightGBM, PyTorch, Agentic AI, MCP, RAG, LangChain

## RELEVANT COURSES

*OUTSTANDING PERFORMANCE (A/A*)

Software Development and Operations*
Statistical Methods and Applications
Fundamentals of Programming
Real Analysis and Linear Algebra
Computer Vision and Deep Learning
Neural Networks & Deep Learning (Online)